

Pharmacological Studies of Artichoke Leaf Extract and Their Health Benefits

Maryem Ben Salem¹ · Hanen Affes^{1,2} · Kamilia Ksouda¹ · Raouia Dhouibi¹ · Zouheir Sahnoun¹ · Serria Hammami¹ · Khaled Mounir Zeghal¹

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Abstract Artichoke (*Cynara scolymus*) leaf extract was one of the few herbal remedies which the clinical and experimental trials have complemented each other. Both experimental and clinical effects have been verified through extensive biomedical herbal remedy research. Specifically, antioxidant, choleretic, hepatoprotective, bile-enhancing and lipid-lowering effects have been demonstrated, which corresponded with its historical use. Ongoing research seems to indicate that artichoke indeed have medicinal qualities. Most significant appears to be its beneficial effect on the liver. In animal studies, liquid extracts of the roots and leaves of artichoke have demonstrated an ability to protect the liver, with possibly even to help liver cells regenerate. Although research is not yet conclusive, scientists were optimistic that its long-standing use in humans for digestive and bowel problems was indeed justified. It may also play a role in lowering cholesterol and thus help to prevent heart disease. Boiled wild artichoke reduced postprandial glycemic and insulinemic responses in normal subjects but has no effect on metabolic syndrome patients. This article intended to review the wide ranging pharmacological effects of artichoke leaf extract.

Keywords Artichoke leaves extract · Cardiovascular disorders · Antioxidant · Hepato-protective

Abbreviations

ALE	Artichaut leaf extract
<i>C.scolymus</i>	<i>Cynara scolymus</i>
LDL-C	Low density lipoprotein-cholesterol
DM	Diabetes mellitus
NASH	Non alcoholic steatohepatitis

Introduction

Cynara was a relatively small genus, originating from the mediterranean area that included two crops, globe artichoke (*Cynara cardunculus* var. *scolymus* L.) and cardoon (*Cynara cardunculus* var. *altilis* DC), as well as their ancestor, wild cardoon (*Cynara cardunculus* L. var. *sylvestris*) as demonstrated [1–3]. The earliest description of the *Cynara* came from the 4th century B.C. Greek writer Theophrastus. These plants were popular with Greeks and Romans as food and in medicine [4, 5].

Globe artichoke was a perennial, rosette plant, widely cultivated for its large fleshy head or capital (immature flowers) which represented 30–40 % of the fresh weight (FW). The edible parts of the artichoke where the tender inner leaves (bracts) and the receptacle commonly known as “heart” constitute nearly 35–55 % of the FW of the head [6]. Besides, there were some non-food industrial by-products such as leaves, external bracts and stems. They represented about 80 % of the biomass and might be used for the extraction of food additives and nutraceuticals [5, 6]. The artichoke crop might last six or more years, reaching the maximum productivity in the third year [6]. Artichoke played an important role in human nutrition, especially in the Mediterranean area [5]. Furthermore, it contributed significantly to the Mediterranean agricultural economy, where more than 60 % of the total world production came from Italy which was classified as the

✉ Hanen Affes
affeshanen13@yahoo.fr

¹ Laboratoire de pharmacologie, Faculté de Medecine Sfax, Sfax, Tunisia

² Laboratory of Pharmacology, Faculty of Medicine of Sfax, Avenue Majida Boulila, 3029 Sfax, Tunisie

leading producer (about 474.000 t annually) followed by Spain (215.000 t), France (55.000 t) and Greece (25.000 t) [5]. Nowadays, the artichoke is cultivated in many parts of the world, such as the United States, mainly in California, in South America (Argentina, Chile, Peru), North Africa, Near East Turkey and Iran and China [3, 5–7]. Artichoke's commercial production is mainly based on perennial cultivation of vegetatively propagate clones [4]. This plant is genetically robust with a marked tolerance to pathogens. This is due to sesquiterpenes, cynaropicrin, which have antifeedant activities. The latest being also responsible for the typical bitter taste of artichoke [6].

Cynara scolymus

Cynara scolymus (Asteraceae) was a native plant of the Mediterranean (North Africa and southern Europe) and popularly known in Brazil as artichoke. Regardless of its origin in the regions of subtropical climates. The *C. scolymus* is cultivated worldwide because of its nutritional benefits and medicinal properties [5]. Since ancient times, it has been cited as a drug in traditional medicine and around the world. It is related to beneficial effects in diseases of the biliary tract, digestive action, and help in the treatment of scurvy, anemia and has antiatherosclerotic effect [8, 9].

It is not only a healthy food known for its pleasant bitter taste, but also an interesting and widespread herbal drug. Artichoke leaf extracts (ALE) are widely used alone or in association with other herbs for embittering alcoholic and soft drinks and also to prepare herbal teas or herbal medicinal products [10]. Polyphenolic compounds presented mainly in the leaves rather than in the artichoke heads have been documented as the active principles of this plant.

Classification

Kingdom	: P l a n t a e
Divison	: Magnoliophyta
Class	: Magnoliopsida
Order	: Asterales
Family	: Asteraceae
Genus	: <i>Cynara</i>

History

Used as a food and a medical remedy as early as the 4th century B.C., artichoke plant has a long history. At the time,

a pupil- of Aristotle named Theophrastus was the first who described the plant in detail. Enjoyed as a delicacy, an appetizer and digestive aid by the aristocracy of the Roman Empire, it later seemed to have fallen into oblivion until the 16th century, when the medicinal use of the artichoke for liver problems and jaundice was recorded.

In 1850, a French physician successfully used an extract of artichoke leaves in the treatment of a boy who got sick with jaundice for a month and did not alleviate from the drugs used at that time. Artichoke is one of the oldest cultivated plants [11]. It was first grown in Ethiopia and then made its way to southern Europe *via* Egypt. Its image was found on ancient Egyptian tablets and sacrificial altars. The ancient Greeks and Romans considered it a valuable digestive aid and reserved it as a rare plant for consumption in elite circles. In sixteenth-century Europe, artichoke was also considered as a “noble” vegetable meant for consumption by the royal and the rich.

In traditional European medicine, the leaves of the artichoke (not flower buds, which were the parts commonly cooked and eaten as a vegetable) were used as a diuretic to stimulate the kidneys and as a “choleretic” to stimulate the flow of bile from the liver and gallbladder.

In the first half of the twentieth century, French scientists began modern research into these traditional medicinal uses of the artichoke plant [11]. Their work suggested that the plant indeed stimulate the kidney and gallbladder. In the mid-century, Italian scientists isolated a compound from artichoke leaf called cynarin, which appeared to duplicate many of the effects of whole artichoke. Synthetic cynarin preparations were used as a drug to stimulate the liver and gallbladder and to treat elevated cholesterol from the 1950s to the 1980s since competition from newer pharmaceuticals had eclipsed the use of cynarin.

Composition of Artichoke Leaf Extract

The chemical components of artichoke leaves had been studied extensively and have been found to be a rich source of polyphenolic compounds including three classes (Figs. 1, 2, and 3) [12, 13].

1. Phenolic acids which include a combination of caffeic acid and quinic acid.

C. scolymus has the important antioxidant compounds; cynarin and chlorogenic acid, by the combination of 1, 3-o-quinic acid with two molecules of caffeic acid 1, 3-di-o-caffeoyl quinic acid (cynarin) and 5- o-caffeoyl quinic acid (cyrogenic acid).

This biological compound of artichoke leaf extract has a low content of fat and high levels of minerals (potassium,

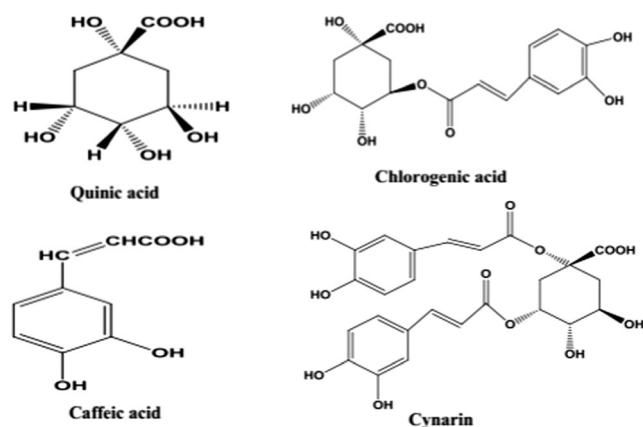


Fig. 1 Structures of phenolic compounds by the combination of quinic acid and caffeic acid [12, 13]

sodium and phosphorus), vitamin C, fibers, polyphenols, flavones, inulin and hydroxycinnamates - caffeoylquinic acid derivatives [7, 14, 15]. According to Foti *et al.* [14], the chemical composition of artichoke seeds is the following: crude protein 21.6 %, crude fiber 17.1 %, crude oil 24.05 % and ash 3.8 %. The artichoke flower heads have a high content of vitamin C (10 mg / 100 g FW) and minerals (K 360 mg / 100 g FW; Ca 50 mg / 100 g FW). Leaves and heads of artichoke have been found to be rich in polyphenolic compounds, inulin, fiber and minerals [5]. ALE has shown antioxidative, antibacterial, anti-HIV, bile-expelling, hepatoprotective, urinate and choleretic activity, as well as the ability to inhibit cholesterol biosynthesis and LDL oxidation [16, 17]. Although artichoke extract has been used as medicine for many years, it has not been extensively examined as antimicrobial agent. Nevertheless, some researchers reported that these extracts have broad inhibitory activities against microorganisms and they could be of promising use in food industry [18, 19]. The exact polyphenol content varies greatly depending on the extraction method [17]. The main phenolic compounds are the caffeic acid derivatives which include the caffeoylquinic acid derivatives. Using the International Union of Pure and Applied Chemistry (IUPAC) nomenclature,

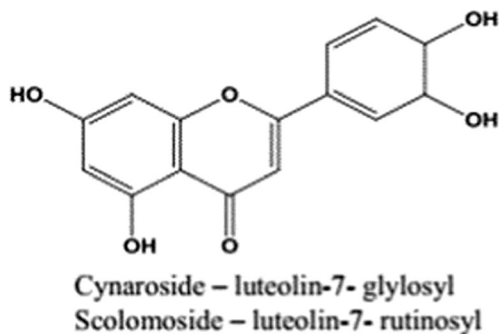


Fig. 2 Structures of flavonoids compounds particularly glycoside of luteolin including of luteoline-7- glycoside (cynaroside) and luteoline-7- rutinoside (sculomoside) [12, 13]

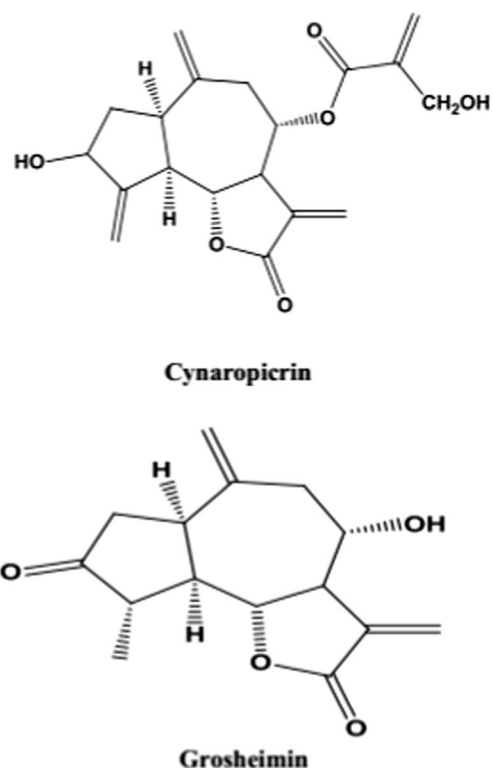


Fig. 3 Chemical structures like sesquiterpin B- selinene and caryphyllene [12, 13]

5-O-caffeoylquinic acid (chlorogenic acid) is the most abundant single substance (39 %), followed by 1,5-O-dicaffeoylquinic acid (21 %) and 3,4-O-dicaffeoylquinic acid (11 %), based on total caffeoylquinic acid content [5, 20]. Furthermore, the 1, 3-O-dicaffeoylquinic acid (cynarin) content in methanolic extracts of artichoke is very low (about 1.5 %), the majority of which is located in the pulp of leaves; although the dried leaves and stems of artichoke also contain this compound [5]. The above phenolic substances have an important scavenging activity against reactive oxygen species (ROS) and free radicals, and perform as a protective shield against oxidative damage to biological molecules, such as proteins, lipids and DNA [14, 21]. In addition, other phenolics such as the flavones 5,7-dihydroxy-2-(4-hydroxyphenyl)-4H-1-benzopyran-4-one (apigenin) and 2-(3,4-dihydroxyphenyl)-5,7-dihydroxy-4-chromenone (luteolin), as well as the anthocyanidins such as 2-(3,4-dihydroxyphenyl) chromenylium-3,5,7-triol (cyanidin), 2-(4-hydroxy-3 methoxyphenyl) chromenylium-3,5,7-triol (peonidin) and 2-(3,4,5-trihydroxyphenyl)chromenylium-3,5,7-triol (delphinidin) have been isolated in only in the heads of artichoke [5, 20]. The above flavones apigenin and luteolin have been identified in leaves and heads of the plant in form of glucosides and rutinosides, whereas anthocyanin pigments are present only in heads in form of glucosides and sophorosides [5].

Role of Active Constituents

ALE contained bioactive and flavonoid compounds such as caffeoylquinic acids and luteolin glucosides. As it is known, cynarin is a major dicaffeoylquinic acid and chlorogenic acid was the main monocaffeoylquinic acid. Whereas luteolin-7-O glucoside is the major flavonoids showed [16, 22]. These compounds have shown several biological activities.

Artichaut Leaves Extract and Health Benefits

In various pharmacological test systems, ALE has been reported to show antioxidative, anti-HIV integrase, liver-protective, bile-expelling, anti-microbial and lipid-lowering effects [9, 16, 22–24]. The mechanisms of these effects are discussed in Fig. 4.

Hypoglycemic Effects

Diabetes mellitus (DM) is a chronic disease characterized by disorders in the metabolism of carbohydrates, lipids and several essential trace elements [25]. The use of lipid lowering diets and drugs are common for treating hyperlipidemia in diabetes [26].

Several reported studies screened some plants having folk medicine reputation for antidiabetic potency [27, 28]. *In vivo* studies demonstrated hypoglycemic effect of ALE. Tables 1 and 2 showed that ALE presented a significant hypoglycemia effects.

Cholesterol Lowering Effects

High cholesterol was associated with an increase risk for coronary heart disease and atherosclerosis. In fact ALE reduced cholesterol levels. The efforts to study this idea dated back all the way to the 1970s, when scientists began examining cynarin and cholesterol in their labs [31]. According to the most recent findings, the constituent luteolin played a crucial role in the inhibiting effect of ALE on cholesterol synthesis [32].

It was obviously acting below the level of HMG-CoA reductase, the key enzyme of cholesterol biosynthesis. In addition, the total extracts showed effects of HMG-CoA reductase, which probably occur *via* other interactions, indirectly *via* activation of inhibitory mechanisms or inhibition of activating mechanisms (Fig. 5). Thus ALE appeared to inhibit lipid synthesis at several levels. Accumulation of undesired sterol compounds described for some synthetic lipid-lowering agents were thus not to be expected with ALE. In addition to findings from animal experiments [33–37], there were many clinical data in the literature proving the lipid-lowering effects of ALE, these effects were confirmed by animals studies (Table 3). In fact these results demonstrated that both caffeoylquinic acids and luteolin glucosides were considered to be responsible for its antiatherogenic actions demonstrated [16, 22, 23, 40]. Also, luteolin was thought to be one of the constituents of ALE which inhibited the novo synthesis of cholesterol [41]. Therefore, in the current study the reduction of atherogenic parameter, total cholesterol and triglyceride levels in treated groups with ALE were due to bioactive and flavonoid compounds of ALE.

Hence artichoke extract was found to have a positive effect on the serum lipid profile values and blood glucose levels [42, 43].

Fig. 4 Mechanisms of artichoke leaf extract [78]

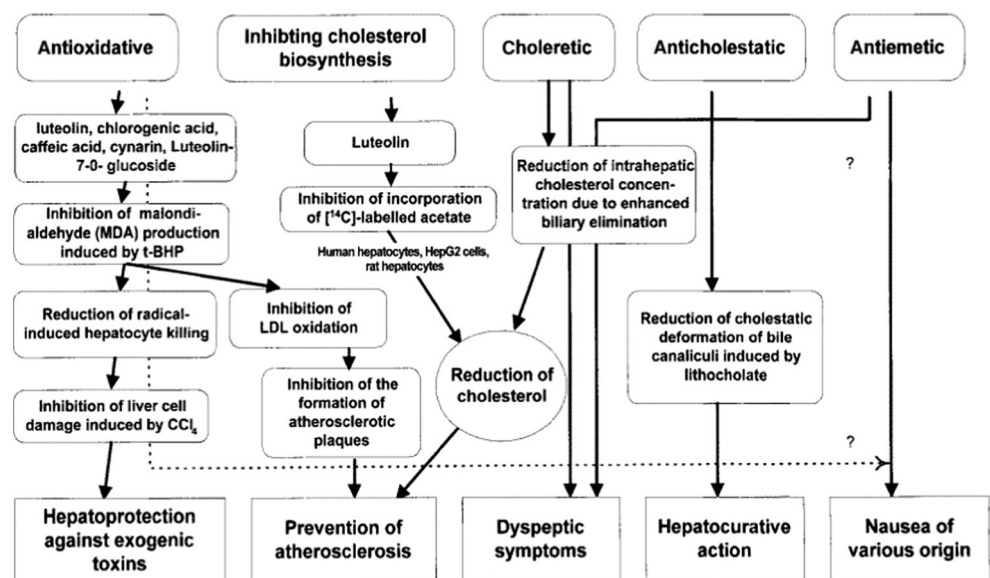


Table 1 Hypoglycemic effect of artichoke leaf extract (ALE) in diabetic humans

Hypoglycemic effect	Type of solvent	Dose (route of administration) (oral)	References
Fasting blood glucose levels : - Decreased from 163.4 ± 4.39 to 138.8 ± 3.9 mg/dl ($P < 0.01$) Postprandial blood glucose levels : - Decreased from 262.22 ± 3.82 to 241.75 ± 3.46 mg/dl ($P < 0.01$).	- Aqueous	- 6 g /kg/day	[29]

Hepatoprotective Effects

The liver is the most vulnerable gland to the toxic effects of xenobiotics, as responsible for their metabolism. Significant impact on the functioning of this gland is related to the lifestyle: alcohol consumption, diet with high fats ingredients and prooxidative substances and synthetic drugs. Very important aspect in herbal medicaments has protective properties on parenchymal organ-damaging. Concomitant intake of plant extracts containing cytoprotective compounds may increase the efficacy of treatment and reduced side effects. One of the plants with hepatoprotective action is artichoke [44].

Many studies demonstrated the hepatoprotective action of ALE, which was related to its active compounds and supported the use of artichoke as a hepatoprotective and antioxidant agent (Table 4). However, more researches were needed here to fully understand the hepatoprotective mechanisms and to reveal the scope of the hepatoprotective effects.

Anti-atherosclerotic Effects

Earlier research on the ability of ALE to promote digestion led to the discovery of its application for cardiovascular health. Researchers noted that artichoke leaf helped reduced cholesterol levels and prevented atherosclerotic deposits.

It was as early as the 1930s when scientists first discovered that artichoke extract had a favorable effect on atherosclerotic plaques in the arteries. Later animal studies, in which rats were fed a high-fat diet, also showed that artichoke extract

prevented a rise in serum cholesterol levels and in manifestations of atherosclerotic plaque [50].

In vitro and laboratory animal data suggested that artichoke extract might have vasodilator activity and provide cardio-protection activities *via* its effect on the regulation of the endothelial nitric oxide synthase (eNOS) gene. This study investigated the effect of ALE on cultured human umbilical vein endothelial cells and demonstrated that treatment with ALE increased activity and expression of the e NOS gene, and increased nitric oxide production. Artichoke extract might offer an additional degree of cardiovascular protection *via* its overall positive effect on endothelial function [50].

The compounds of ALE such as cynarin might prevent the development of atherosclerotic plaques, some of these studies used rats, which were a poor model for human cholesterol metabolism. The anti-atherosclerosis action was thought to be the product of two mechanisms of action: an antioxidant effect that reduced lipoprotein low density (LDL) oxidation and inhibition of cholesterol synthesis [11].

Some researchers have shown that dyslipidemia is one of the major cardiovascular risk factors. It has been suggested that insulin deficiency in DM is associated with a variety of abnormalities in metabolic and regulatory processes, which caused the accumulation of lipids such as total cholesterol and triglyceride in diabetic patients [51].

In addition to findings in animal experiments, a number of early case reports and uncontrolled studies indicated clinical effectiveness of the artichoke extract on human cholesterol levels [52].

Table 2 Hypoglycemic effect of artichoke leaf extract (ALE) in diabetic rats

Hypoglycemic effects	Type of solvent	Dose (route of administration) (oral)	References
Streptozocotin-induced diabetic rats: - Decreased glucose level in treated diabetic rats as compared with diabetic control rats ($P < 0.001$)	- Aqueous	- 200 mg/kg/day and 400 mg/kg/day	[40]
Alloxan –induced diabetic rats : - Decreased glucose level in treated diabetic rats as compared with diabetic control rats ($p < 0.05$)	- Hydro alcoholic	- 300 mg/kg/day	[30]

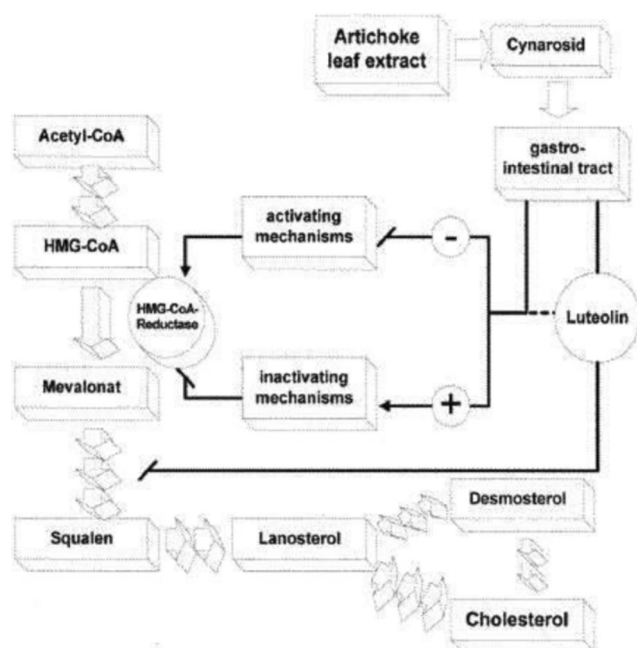


Fig. 5 Inhibition of cholesterol synthesis by artichoke leaf extract [76]

The discovery that ALE reduced cholesterol levels opened up exciting perspectives in the prevention and treatment of arteriosclerosis and coronary heart diseases.

Antioxidant Effects

Oxidative stress was defined as an imbalance between local reactive oxygen species production (chiefly superoxide anions, hydrogen peroxide, hydroxyl radicals) and antioxidative mechanisms. The protective effect of flavonoids against oxidative injury might depend not only on their antioxidant activity, but also on their affinity and permeability through plasma membranes [16].

In vitro studies have isolated a variety of antioxidant compounds from globe artichoke, and a confirmation of the antioxidant activity of artichoke has been noted in multiple pre-clinical studies [16, 22, 36, 53–56] (Table 4).

Further studies have confirmed an antioxidant-related hepatoprotective effect *in vitro* on rat hepatocytes, cultured endothelial cells and monocytes, and *in vivo* rat model [34–36, 53, 56].

An additional study demonstrated further antioxidant properties *via* inhibition of LDL oxidation [57]. ALE is known to have antioxidant potential, many *in vitro* studies have

demonstrated that the antioxidant potential of ALE is due to radical scavenging and metal ion chelating effect of its constituents such as cynarin, chlorogenic acid and flavonoids [57, 58].

Prebiotic/Probiotic Effects

An *in vitro* study found that inulin extracted from globe artichoke is a potent probiotic stimulator of beneficial *Bifidobacterium bifidum* [59]. This result investigated the ability of artichoke to serve as vehicle for the delivery of probiotic bacterial strains.

Choleretic Effects

In vitro studies have demonstrated that ALE might increase bile secretion in perfused rat liver and liver cell cultures [9, 60]. It has been suggested that the reduction of intrahepatic cholesterol concentration is responsible for globe artichoke extract's ability to treat dyspepsia.

Others studies showed a significant increase in bile flow that was observed after acute treatment with ALE as well as after repeated administration. The choleretic effects of ALE was similar to those of the reference compound dehydrocholic acid. Total bile acids, cholesterol and phospholipids were determined by enzymatic assays. There was a strong ALE-induced increase in total bile acid concentration over the entire experiment. With the highest dose (400 mg/kg), a significant increase was obtained after single and repeated administration. The bile acids increased effects of ALE were much more pronounced than those of reference dehydrocholic acid.

Genotoxicity Effects

A micronucleus test is a test used in toxicological screening for potential genotoxic compounds. There are two major versions of this test, one *in vivo* and the other *in vitro*. *In vivo* test normally uses mouse bone marrow or mouse peripheral blood. When a bone marrow erythroblast developed into a polychromatic erythrocyte, the main nucleus is extruded; any micronucleus that has been formed may remain behind in the other enucleated cytoplasm. Visualization of micronuclei is facilitated in these cells because they lack a main nucleus. An

Table 3 Hypolipidemic effects of artichoke Leaf extract (ALE) in diabetic animals

Hypolipidemic effects	Type of solvent	Dose (route of administration)	References
- Reduction in total cholesterol of 4.2 %	-Aqueous	- 1280 mg Over (12 week)	[38]
- 18.5 % reduction in cholesterol level	-Aqueous	- 1280 mg Over (6 weeks)	[39]

Table 4 Phenolic compounds of Artichaut Leaf Extract (ALE) and their biological activities

Majors phenolic compounds of ALE	Biological activities	References
- Luteolin 7-O-glucoside - Cynarin - Acid chlorogenic - Quinic acid - Caffeic acid	Antimicrobial and antifungal activity: -relatively higher activities against a <i>Bacillus subtilis</i> , <i>Escherichia coli</i> , <i>Pseudomonas aeruginosa</i> , <i>Staphylococcus aureus</i> , and four molds, <i>Aspergillus niger</i> , <i>P. oxalicum</i> , <i>M. mucedo</i> , and <i>Candida. cucumerinum</i> , whereas they showed weak activity against <i>Sacchro. cereVisiae</i> . Hypocholesterolemic effects: -antiatherogenic actions and inhibition <i>de novo</i> synthesis of cholesterol, triglyceride and total cholesterol levels in treated groups with ALE . These indicate that ALE has a lipid lowering effect on the diabetic rats. -blocked the stimulation of cholesterol biosynthesis by insulin. Antioxidant activity: - showed significant decrease of plasma MDA (Plasma malondialdehyde) levels ($p < 0.01$) as compared to the OX STRESS animals. - a significant impact on 8-OHdG (urinary 8-hydroxydeoxyguanosine), and GSH levels (erythrocyte reduced glutathione) and also significantly decreased plasma FRAP (ferric reducing antioxidant power) values and protein oxidation in plasma and erythrocytes, and an increased activity of the antioxidant enzyme GSH peroxidase in erythrocytes Hepatoprotective effects: - help to remove dangerous toxins and digest fats by bile production of the liver - averted the process of lipid peroxidation from occurring in the cell membranes of the tissues of the liver, making the artichoke an ideal weapon in the arsenal to help to obtain optimal liver function. - cell protective effect when studying rat liver cells exposed to t-BHP (tertiary butylhydroperoxide), found that ALE significantly prevented oxidative damage to hepatocyte membranes Other activities: - immunomodulatory, anticarcinogenic effects - prevention of cardiovascular diseases - gastro intestinal tract effects	[45, 62] [16, 22, 23] [46, 58–60] [23–44, 47–49, 72] [48]

increase in the frequency of micro nucleated polychromatic erythrocytes in treating animals is an indication of induced chromosome damage.

Previous studies have reported the potential genotoxicity of some traditional medicines [61, 62]. Assessment of the genotoxic properties of folk medicine is an important topic, since damage to the genetic material may lead to critical mutations and therefore also an increased risk of different diseases, including cancer [61].

Several investigations on ALE have demonstrated its efficacy in genotoxic effects using the comet assay and micronucleus test in mice.

Other studies show that the potential genotoxic effects of methanolic extracts of *Euphorbia hirta*, which is commonly used in traditional medicine to treat a variety of disease conditions including asthma, coughs, diarrhea and dysentery, are investigated using *Allium cepa* assay, demonstrating that the methanol extracts of *E. hirta* exert significant genotoxic and mitodepressive effects at 1000 µg/ml [61].

Recent studies have investigated genotoxicity of ALE. Both genotoxic and antigenotoxic effects were observed depending on the concentrations investigated, high doses of ALE (5 mg/ml and 2 g/kg in cell line and mice studies, respectively) showed

mutagenic effects, while lower doses showed protective antioxidant effects [63].

Cancer Prevention

There are many other powerful polyphenol-type antioxidants found in artichoke that could contribute to the prevention and management of prostate cancer, breast cancer, and leukemia. Studies have shown that antioxidants rutin, quercetin, and gallic acid found in ALE are able to induce apoptosis (cell death) and reduce the proliferation of cancer cells.

In research done at Comenius University in Slovakia, ALE was studied for its ability to inhibit growth of leukemia cells. Over a 24-h period, leukemia cells were treated with a variety of concentrations of ALE, with results suggesting that it exerted an antiproliferative activity on leukemia cells while inducing apoptosis of these cells as well [64]. Furthermore, researchers at the University of Georg-August in Germany indicated that the many phytochemicals in artichoke helped to block secretion of cancer agents, thus inhibiting angiogenesis related to cancer.

Anti-inflammatory Activity

Non-alcoholic fatty liver disease is a clinical pathologic entity increasingly recognized as a major health burden in developed as well as in developing countries. It includes a spectrum of liver damage ranging from simple steatosis to non-alcoholic steatohepatitis, advanced fibrosis, and probable progression to cirrhosis [65].

Among the many causative factors of non-alcoholic NASH, oxidative stress, lipid peroxidation and inflammation were considered the most probable risk factors [66] (Fig. 6).

Results showed that ALE revealed a significant increase in parameters of anti-inflammatory indicating that serum NF- κ B, TNF- α , Cox-2, CD 40 and HGF levels have been increased significantly as a consequence of treatment with either (crude aqueous methanolic extract) or (crude aqueous fraction) as comparing to the untreated NASH-induced rats.

Histological studies of liver tissues of steatohepatitis - induced group treated with CSM, showed significant reduction in fatty infiltration as compared with that in the untreated NASH-induced group (Fig. 6).

Digestive Effects

The high concentration of cynarin in ALE not only affects cholesterol, but also could contribute to aiding in digestive health. Cynarin is known to stimulate the production of bile, which enables to digest fats and absorb vitamins from our food, making artichoke an excellent way to start any meal. Studies have shown that ALE could be very helpful for people suffering from irritable bowel syndrome and dyspepsia, or upset stomachs. In a study done at the University of Reading in the United Kingdom, 208 adults who suffered from IBS and dyspepsia were monitored over a 2-month period of intervention with ALE. Results showed a 26.4 % reduction in Bowel syndrome incidence among the participants at the end of the trial. A significant shift of self-reported bowel patterns away from “alternating constipation/diarrhea” toward “normal” were observed as well. Dyspepsia symptoms also decreased

by 41 % after treatment, and in general, the participants noted a 20 % increase in quality of life after treatment [32].

Cardiovascular Health Effects

Early research on the ability of ALE to promote digestion led to the discovery of its application for cardiovascular health. Researchers noted that artichoke leaf helps to reduce cholesterol levels and prevent atherosclerotic deposits. More recent research confirms these findings. In one 12-week, double-blind study, ALE was found to decrease cholesterol levels significantly in those who had high (more than 220 mg/dl) cholesterol levels, ALE had a tendency to increase HDL (good) cholesterol levels. The study also found that those subjects with the highest beginning cholesterol levels showed the most significant change [85].

A 6-week study of more than 300 subjects showed an average reduction in serum cholesterol of nearly 12 %, and a reduction in serum triglycerides of nearly 13 % [9].

The discovery that ALE reduced the elevated cholesterol levels opens up exciting perspectives in the prevention and treatment of arteriosclerosis and coronary heart disease.

It was as early as the 1930s that scientists first discovered that artichoke extract had a favorable effect on atherosclerotic plaques in the arteries. Later animal studies, in which rats were fed a high-fat diet, also showed that artichoke extract prevented a rise in serum cholesterol levels and the manifestation of atherosclerotic plaque. In addition to findings in animal experiments, a number of early case reports and uncontrolled studies indicated clinical effectiveness of the artichoke extract on human cholesterol levels.

Recent research confirms these earlier findings. The above mentioned study by Fintelmann [72] demonstrated a significant reduction in cholesterol and triglyceride levels in spite of the relatively short duration of the study (6 weeks). On an average there was an 11.5 % reduction in serum cholesterol, from 264 to 234 mg/dl. Serum triglycerides were similarly reduced from 215 to 188 mg/dl, corresponding to a decrease of 12.5 %. Although this was an open study, its reliability was

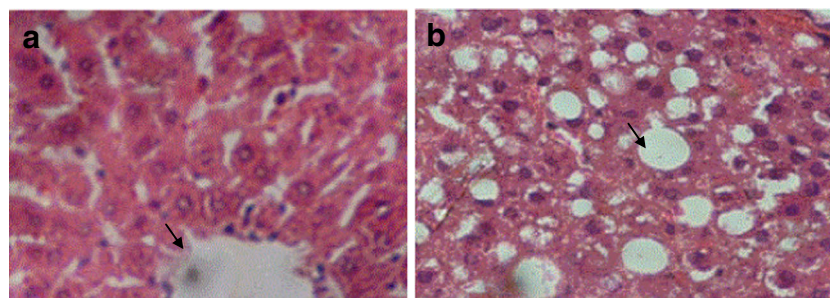


Fig. 6 **a** Liver section of NASH induced rat showing a high degree of hepatocellular cytoplasmic vacuolation (macrovesicular and microvesicular steatosis). **b** Liver section of NASH-induced rat treated

with CSM showing significant reduction in fat deposits in liver tissues evaluated by haematoxylin and eosin (H&E) staining (100 $\times$) [67]

buttressed by the relatively large number of patients (302) and the very high level of statistical significance attained for the main results.

Very interesting results came out of an excellent double blind clinical trial [85], it studied the cholesterol-lowering effect of ALE on 44 healthy individuals under strictly controlled conditions over a 12-week period. There was a significant decrease of cholesterol levels in the volunteers who had high initial levels (greater than 220 mg/dl). In fact, the higher the initial cholesterol value, the higher significant the reduction in cholesterol levels. It was also observed that the protective HDL cholesterol levels showed a tendency to increase.

Although the cholesterol-lowering effect of artichoke extract has been known for a few decades, the mechanism has not been clear. Current research has increased our understanding in this respect.

Artichoke extract has been found to affect the cholesterol metabolism in two different ways. It not only increase the breakdown of cholesterol to bile salts and enhance their elimination through increased bile production and flow; it also inhibits the internal production of cholesterol in the liver.

The inhibiting effect of ALE on cholesterol synthesis was demonstrated in very interesting studies by Gebhardt [26] on rat hepatocyte (liver cells). A highly significant concentration-dependent inhibition of cholesterol synthesis was found. The study indicated that ALE reduced the formation of cholesterol in a physiologically favorable, long-lasting manner. This reduction of cholesterol synthesis persisted for some hours following the period of exposure.

The study further indicated that artichoke extract may work through indirect inhibition of the enzyme HMG CoA-reductase, which might avoid problems known to occur with strong direct inhibitors of HMG CoA-reductase during long-term treatment. The indirect inhibition was supported by the fact that ALE effectively blocked insulin-dependent stimulation of HMG CoA-reductase without affecting insulin in general.

HMGCoA-reductase is a key enzyme in cholesterol synthesis, and HMG CoA-reductase inhibitors generally reduce total cholesterol, LDL -cholesterol and triglyceride [11, 21, 68, 69].

Pharmacodynamics/Kinetics

The active constituents of ALE have not been detected in human plasma or urine. Several metabolites detected in human plasma have been identified as being derived from mono and dicaffeoylquinic acids, and from flavanoids that are known active phenolic constituents of ALE. The metabolites caffeic acid, ferulic acid, and isoferulic acid achieved peak plasma concentrations within 1 h and declined over 24 h, demonstrating a near biphasic profile. The hydrogenated metabolites dihydrocaffeic acid and dihydroferulic acid were detected after 6 to 7 h, suggesting the involvement of more than

one pathway in processing caffeoylquinic acids. Peak plasma concentration of luteolin was reached within 30 min, with elimination showing a biphasic profile [70].

Gastro Intestinal Tract Effects

Because of its long historical use for liver conditions it seemed reasonable to investigate the artichoke plant scientifically, and the first clinical studies were conducted in the 1930s with encouraging results. In the 1990s the interest was intensified, and several excellent clinical studies have been conducted during the last few years: realizing the importance of adequate bile flow for health, German researchers set out to confirm the earlier findings of bile promoting effect of the artichoke plant in a controlled double-blind study on healthy volunteers [71]. The participants were given an onetime dose of artichoke extract or placebo, and their bile secretion was measured with special techniques over the following hours. The bile secretion was found to be significantly higher in the group that received the artichoke extract. Another clinical study showed an improvement of symptoms in 50 % of patients with dyspeptic syndrome after 14 days of treatment with artichoke leaf extract.

Interesting results were also demonstrated in a large open label study of 417 participants with liver or bile duct disease. Most of these patients had longstanding symptoms, some of them for many years. They suffered from upper abdominal pain, bloating, constipation, lack of appetite and nausea. These patients were treated with artichoke leaf extract for 4 weeks. After 1 week, around 70 % of the patients experienced improvement of their symptoms, and after 4 weeks the percentage was even higher (approx. 85 %). Even more remarkable improvement was shown in another recently completed open label study, where 553 out patients with non specific dyspeptic complaints were treated with a standardized artichoke leaf extract. The subjective complaints declined significantly within 6 weeks of treatment [72]. Improvements were found for vomiting (88 %), nausea (83 %) and abdominal pain (76 %), loss of appetite (72 %), severe constipation (71 %), flatulence (68 %) and fat intolerance (59 %). Ninety-eight percent of the patients judged the effect of the extract to be considerably better, somewhat better or equal to that achieved during previous treatment with other drugs. The dosage used in this study was 1–2 capsules three times daily of the preparation Hepar SL Forte. One capsule contained 320 mg of dry extract of artichoke leaves, standardized to provide 3 % of caffeoylquinic acid [72].

The study by Fintelmann [72] not only confirmed the efficacy of the artichoke extract for dyspepsia, but also demonstrated a significant effect of the extract on fat (lipid) metabolism. The researchers found a significant decline in both the cholesterol

and triglyceride levels in the blood, which confirmed a discovery, made as early as in the 1930s [7, 14, 15].

Availability

New ALE was now available in the United States, giving Americans a chance to discover its merits. While the German artichoke products cited in most European studies, typically contain 3 % caffeoylquinic acids. This new artichoke extract was standardized to contain 15 % caffeoylquinic acids, calculated as chlorogenic acid [73].

The Use of Cynara Crops on Ruminant Nutrition

Ensilage of artichoke by-products remaining after industrial processing is an economical and environmentally friendly way to decrease waste discharges and reduce the management cost. According to Meneses et al. [74], these products are suitable for ensiling, with pleasant smell and good silage characteristics. Such silage could be used in ruminant feeding, replacing conventional roughage sources such as hay [75]. The use of artichoke bracts in lamb diets resulted in higher percentage of omega-3 fatty acids in fat deposits [76]. When artichoke silage was fed to dairy ewes, no difference was observed in milk composition, while the manufactured cheese had lower fat content and total free fatty acids than controls [77]. According to Sallam et al. [78], green forage, dry leaves and stalks of cardoon had high nutritive value when they were fed to adult wethers. In particular, the forage had very high digestibility for organic matter (86 %), while ensilage is the most appropriate way for preserving it for long periods. In another research, whole cardoon seeds were included in wethers' diets and it was concluded that the most suitable level seemed to be around 10 % in order to take advantage of the digestive effects. The above researchers reported that adding more than 10 % seed to ruminant diets should be avoided in high productive animals, as the high fiber content of the seed could limit feed intake and the very low content of under graded protein could decrease productivity.

Potential Health Benefits

The polyphenolic constituents of artichoke extract are by now widely recognized to be powerful antioxidants. One of the caffeoylquinic derivatives in particular, chlorogenic acid, has been repeatedly investigated during the last couple of years, with interesting results pointing to new territories, such as HIV, cancer, glucose metabolism and more.

Chlorogenic acid was tested in a study of chemically induced precursor lesions to colorectal cancer in rats. Significant

results were achieved both in preventing and reducing these lesions in the group that was treated with chlorogenic acid.

An investigation on HIV replication in tissue cultures demonstrated that caffeoylquinic acids are a potentially important class of HIV inhibitors that could contribute to our understanding of the mechanisms for viral integration into the host cells. These compounds act as a site distinct from that of current HIV therapeutic agents and are promising leads to new anti-HIV therapeutics. An important observation in this study was also that caffeoylquinic acids are effective against the virus at only one hundredth concentrations at which they exhibit toxicity.

Another potential field for further exploration is the glucose metabolism in the liver. A study from Germany in 1998 found chlorogenic acid to be an effective inhibitor of the so-called hepatic glucose-6-phosphatase system, which regulates blood glucose levels. Such a Gl-6-P inhibitor may be useful for the reduction of inappropriately high rates of glucose output from the liver, which was often found in non-insulin-dependent diabetes.

Numerous animal studies have indicated that chlorogenic acid is effective in inhibiting carcinogenic reactions, and played an integral role in modulating the carcinogenic potential of toxic chemicals.

These results might suggest possible future application of this natural substance to chemoprevention of cancer.

One particularly interesting feature of chlorogenic acid is its antioxidant effect against a substance called peroxynitrite. It has been shown, for example, that chlorogenic acid prevents oxidative damage to DNA by scavenging peroxynitrite and it is a cytotoxic agent that forms in the body from the reaction between superoxide and nitric oxide. Through recent research it was established that peroxynitrite is one of the major damaging oxidants produced in humans.

As our tissues are continually exposed to damaging "reactive oxygen species" or free radicals, it is important to have effective defense mechanisms to protect against or repair the damage caused by these free radicals. Major defense mechanisms include certain enzymes, such as superoxide dismutase, and antioxidants such as vitamin C, vitamin E and glutathione. Oxidative stress occurs when the production of damaging radicals overwhelms the antioxidant defenses. Chlorogenic acid may prove to add substantial force to these defenses. Peroxynitrate as well as chlorogenic acid will most likely remain a focus of research for many years to come. Chlorogenic acid in artichoke leaf has been identified as a potent antioxidant, and researchers are examining its possible application in treating cancer, diabetes and HIV. Chlorogenic acid has also been shown to be an antioxidant against peroxynitrite, one of the most damaging oxidants, and one that is associated with heart disease, neurodegenerative diseases and inflammatory disease.

In one study, chlorogenic acid was shown to both prevent and reduce existing precancerous lesions in the colons of rats. Other studies have suggested that chlorogenic acid helps inhibit carcinogenic reactions and lessens the potential of certain chemicals to be carcinogenic.

A study on the application of artichoke leaf for diabetes found that chlorogenic acid helps regulate glucose metabolism, preventing the risk of high rates of glucose output found in diabetic patients.

A study on HIV replication showed that caffeoylquinic acids have promise as HIV inhibitors, and that they may be effective against the virus at only 1/100th the concentration at which they exhibit toxicity [72].

Applications of Globe Artichoke

Artichoke could be eaten as a fresh, canned or frozen vegetable [5, 79]. Historically, this plant has been used in folk medicine since Roman times, for its health benefits which were mainly due to the high content of polyphenols and inulin [5, 7, 15]. These substances are very important for the human nutrition since they are involved in the prevention of cancer [80, 81]. Among the common edible plants, artichoke is the richest source of dietary antioxidants [82], therefore it could be used in phytopharmaceutical applications [5, 14]. The pharmacologic properties of artichoke flower heads are well documented in several *in vivo* and *in vitro* studies for the treatment of hepato-biliary dysfunction, dyspeptic syndromes, gastric diseases, as well as for inhibition of cholesterol biosynthesis and low density lipoproteins (LDL) oxidation – agents responsible for arteriosclerosis and coronary health disease [5, 14]. ALE decreased serum lipids, as well as hepatic and cardiac oxidative stress in rats fed on high cholesterol diet [83]. Also, wild artichoke extracts fed to aged rats seemed to exert cardioprotective effects [55]. According to the European Medicines Agency (2011) the pharmaceutical forms of artichoke are acceptable with respect to clinical safety. Apart from the edible parts of artichoke, the agro-industrial by-products such as leaves, external bracts and stems which represent about the 80 % of the plant biomass could be a promising and cheap source of healthpromoting polyphenolic compounds and inulin [5, 15]. According to some researchers, water extract of artichoke was reported to give protection against the suppressive effects of ochratoxin on egg production of laying hens or on humoral immune response in broiler chicks [84, 85].

Tolerability and Contraindications

ALE is well tolerated and has few side effects in recommended dosages. The use of the artichoke plant as food in many

countries over hundreds of years supports its safety. However, it is important to note that several rigorous studies report the absence of adverse effects when using a standardized extract compared to placebo. In a large safety study, only one out of 100 subjects reported mild side effects such as transient increase in flatulence.

Local eczematous reactions have been reported after occupational exposure and skin contact with the fresh plant or its dried parts. Such an allergy should be considered a contraindication for internal use of the extract, although no reactions to orally ingested extract have been observed so far. Because of its bile stimulating effect, the extract should not be taken by individuals with gallstones or other bile duct occlusion [68].

Conclusions

This review evaluates the preventative effects of artichoke leaf extract; some researchers have showed a significant decrease in levels of biological parameters in hematological tests, biochemical (glucose levels, cholesterol rate, atherogenic index) and cardiovascular disorders.

These results show that ALE is one of the safest herbal drugs with almost no adverse effects in most of the subjects.

Human studies show that ALE has not been associated with significant side effects in studies so far, but full safety testing has not been completed. They indicate that ALE should not be used by pregnant or nursing women. Safety in young children or in people with severe liver or kidney disease has also not been established.

For all these results, ALE has been proven to be a safe and natural way to maintain and improve general health, because of its many applications for essential physiological functions. As a nutritional supplement and antioxidant, it could safely be used as an adjunct to conventional therapies.

For this review, we allow to make different metabolic tests (*i.e.*, diabetes, cardiovascular) *in vivo* to evaluate the beneficial effects of artichoke leaf to human health.

Conflict of Interest The authors declare that they have no conflict of interest.

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